Deyu Rao

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6072 LSK Business Building, Department of Economics
HKUST, Clear Water Bay, Hong Kong SAR

Academic Appointment

Assistant Professor, Department of Economics, Hong Kong University of Science and Technology
HONG KONG SAR, *July 2022 -*

Faculty Associate, Center for Economic Policy *October 2022 -*
Institute for Emerging Market Studies *September 2024 -*
IAS Center for AI for Scientific Discoveries *January 2026 -*
Center for Economic Growth *May 2026 -*

Cross-Campus Faculty Affiliate, Hong Kong University of Science and Technology (Guangzhou)
GUANGZHOU, *September 2023 -*

Education

Ph.D., Applied Economics and Management, Cornell University ITHACA, *2022*
M.Sc., Applied Economics and Management, Cornell University ITHACA, *2017*
B.Econ. with Distinction, Energy Economics, Renmin University of China BEIJING, *2015*
Visiting Student, Department of Economics, University of Warwick COVENTRY, *2013*
B.A., English Language and Literature, Renmin University of China BEIJING, *2015*

Research and Teaching Interests

Environmental Economics, Urban Economics, Applied Microeconomics

Research

Publications

Research Papers:

Panle Jia Barwick, Shanjun Li **Deyu Rao**, and Nahim Bin Zahur. The Healthcare Cost of Air Pollution: Evidence from the World' s Largest Payment Network. *Review of Economics and Statistics*, 108(1), 2026: 110-128. DOI: 10.1162/rest_a_01430.

Shipei Zeng and **Deyu Rao**. Random Forests with Economic Roots: Explaining Machine Learning in Hedonic Imputation. *Computational Economics*, 66(3), 2025: 2457–2481. DOI: 10.1007/s10614-024-10798-9.

Book Chapters:

Chapter 1, *Trends in and Causes of Global Freshwater Decline*, and
Chapter 2, *Cascading Effects: The Impact of Continental Drying on People, Food, and the Planet*, in Fan Zhang, Christian Borja-Vega, Hrishikesh Arvind Chandanpurkar, James Famiglietti, Rick Hogeboom, Regassa Namara, Zarif Rasul, Pavel Luengas-Sierra, and **Deyu Rao**. 2025. *Continental Drying: A Threat to Our Common Future*. *Global Water Monitoring.* Washington, DC: World Bank. DOI: 10.1596/978-1-4648-2269-8.

Working Papers

Deyu Rao, Zeyu Chen, Mengwei Lin, and Lin Yang. Pollution in the Administrative Calculus: Evidence from Land Allocation in China. 2026.

Winner, Best Paper Award, CAERE Annual Conference NANJING, *May 2026*
Finalist, Best Paper Award, Conf. on Asia and Pacific Economies (CAPE) SUZHOU, *May 2026*

Zeyu Chen, **Deyu Rao**, Lin Yang, Wen Wang, and Yatang Lin. Passenger Travel Costs and the Spatial Organization of Economic Activity.

Deyu Rao. The Role of Environmental Amenities in the Urban Economy: Evidence from a Spatial General Equilibrium Approach. 2023. Under review.

Panle Jia Barwick, Dave Donaldson, Shanjun Li, Yatang Lin, and **Deyu Rao**. Transportation Networks, Short-term Mobility, and Pollution Exposure: Evidence from High-Speed Rail in China. NBER Working Paper No. 30462, 2024. Reject and resubmit, *AER: Insight*.

Deyu Rao, Shanjun Li, Ivan Rudik, and Fan Zhang. Water Woes in a Woven World: The Economic and Trade Impacts of Water Stress. 2024.

Wangyang Lai, Shanjun Li, **Deyu Rao**, Yongfan Zhao, and Fan Zhang. The Impact of Water Resources on Trade under a Changing Climate. 2024.

Work-in-Progress

Deyu Rao, Yuta Suzuki, Sifan Xue, Yun Liu, and Tao Tang. Trade War "Winners" Revisited: Environmental Consequences of Manufacturing Relocation.

Deyu Rao and Yiyuan Zhou. Bridging Divides: The Influence of Urban Transportation on Economic Equity.

Yan Zhang, Shiyu Wang, and **Deyu Rao**. Do Local Government Officials Tunnel Public Goods to Corporations in China? Evidence from Subway Land Auctions.

Zhengyuan Wu, **Deyu Rao**, and Shanjun Li. Quantifying the Impact of COVID-19 on Economic Activities Through Air Pollution.

Nahim Bin Zahur and **Deyu Rao**. Traffic Congestion and Consumer Spending.

Fellowships, Grants, and Honors

PI, General Research Fund (GRF), Hong Kong Research Grants Council *Jan 2027 - Dec 2029*

HK$546,452, "Climate's New Toll: The Economic Costs of Extreme Weather on Transportation Systems", Co-I: Eric Yongchen Zou

Participant, National Natural Science Foundation of China (NSFC), General Program *Jan 2027 - Dec 2030*

RMB 400,000, PI: Yuanning Liang, "How Local Government Capacity Shapes Environmental Performance and Ecological Restoration: A Multi-level Analysis Based on Legislative Decentralization Reform"

PI, General Research Fund (GRF), Hong Kong Research Grants Council *July 2025 - June 2028*

HK$827,500, "Bridging Divides: The Influence of Urban Transport'n on Economic Equity", Co-I: Yiyuan Zhou

PI, Early Career Scheme (ECS), Hong Kong Research Grants Council *Jan 2024 - Dec 2025*

HK$535,443, "Local Governance and Environmental Regulation: Evidence from China's Urbanization"

PI, Research Grant from Li & Fung Supply Chain Institute *Sep 2025 - Aug 2027*

HK$80,000, Co-I's: Yuta Suzuki and Sifan Xue

PI, Research Grant from Institute for Emerging Market Studies (IEMS) *June 2025 - May 2027*

HK$80,000, Co-I's: Yuta Suzuki and Sifan Xue

Teaching Experience **adj. mean [enrollment]**

		2023SP	2024SP	2024FA 2025SU	2026SP	2027SP
ECON 2103	Principles of Microeconomics		4.1 [57]	4.6^{+}[50]	4.7 [68]	*
ECON 4454[1]	Green Economy and Sustainability	4.6^{+}[31]	4.7 [17]	4.7^{+}[52]	4.4 [45]	*
ECON 5670[2]	Urban Economics	4.9 [11]	5.0^{+}[46]	5.0^{+}[30]	4.8 [57]	*

[1]Previously ECON 4999X [2]Previously ECON 6100Q *Scheduled

$^{+}$Recognition of Excellent Teaching Performance

Finalist, The Franklin Prize for Teaching Excellence 2025, HKUST SBM *May 2026*

Professional Activities * scheduled

Invited Seminars and Conference Presentation:

2026: CUHKSZ HSS, HKUST CEP, Fudan CCES, CAPE XJTLU, CAERE, SJTU ICCDS, SUFE, AMES, PKU CCERSI, HKUST-Fudan-SMU Trade, Jinan GLO, SEU Nanjing Urban, SMUE Wuhan U, Hitotsubashi

2025: Tsinghua SEM, Duke Kunshan, HKUSTGZ

2024: SUFE, AERE Summer, RUSE Nanjing, HKU Beijing & Tsinghua CIDEG

2023: ASSA-AREUEA, HKUST CEP, SJTU Antai, PKU, Jinan IESR

2022: HKUST, HKU, NUS, ICCDS, AsRES-AREUEA, UEA, TWEEDS

2021: Cornell, Renmin U, NTU-CEC Workshop

2017-2020: Cornell (CICER & EMI Research Day, AEP, SEERE, Emerging Markets PhD Research Day), NBER Spring Meeting 2017

Discussant:

2026: CAPE, SEU; *2024*: SUFE; *2022*: ICCDS, AsRES-AREUEA, UEA

Professional Services

Referee:

Economic Journal, Review of Economics and Statistics, Management Science, Journal of the European Economic Association, Journal of Development Economics, Journal of the Association of Environmental and Resource Economists, Journal of Environmental Economics and Management, Regional Science and Urban Economics, Labour Economics, Journal of Economic Behavior and Organization, Journal of Applied Econometrics, Environmental and Resource Economics, China Economic Review, Journal of Transport Economics and Policy, Journal of Policy Analysis and Management, Journal of Risk and Insurance, European Journal of Finance, Energy Journal

Research Committee, UG Committee, Honor Thesis Reviewer, etc., Dept. of Economics, HKUST *2023 -*

Part-time Consultant, World Bank Group *2022 - 2025*

Judging Panel Member, Sustainable Smart Campus (SSC) Student Competition, HKUST *2023*

Judging Panel Member, Hong Kong Economic Policy Challenge (HKEPC) *2025*

Seminar Committee, Graduate Student Assembly (GSA) of AEM, Cornell University *2018-2019*

Selected Press Coverage

"World Annual Fresh Water Losses Could Supply 280 Million People, World Bank, Nov 2025.

"High-Speed Rail Enables Chinese to Avoid Unhealthy Environs," NBER Digest, Jan 2023.

"How Smog Affects Spending in China," The Economist, June 2018.

"The Impact of Air Pollution on Healthcare Spending in China," etc., in VoxDev (Apr 2019), VoxEU (Sept 2018), VoxChina (Aug 2017), Cornell Econ News (Aug 2018).

Others: ChinaFund (Jan 2024) on ESG Standards.

Previous Teaching and Research Experience

Best Ph.D. Dissertation Award, Dyson School of Applied Economics and Management, Cornell University *2023*

Outstanding Graduate Teaching Assistant, College of Agriculture and Life Sciences, Cornell University *2021*

Graduate Fellowship, Summer Graduate Research Assistantship, Cornell University *2017-2021*

Scholarship for Academic Excellence, Sports Excellence, Graduate with Distinction, Renmin U of China *2012-2015*

Teaching Assistant

Econometrics, AEM 7100 (PhD level), Prof. Shanjun Li *Spring 2020, 2021, 2022*

Investments, AEM 4670 (UG level), Prof. Sumudu Watugala *Fall 2019, 2020*

CICER Summer Camp (Market Simulation, Marketing, Behavioral Economics) *2019-2021*

Research Assistant

Prof. Shanjun Li, Cornell University Ithaca, *2016, 2018, 2021*

Cornell Institute for China Economic Research (CICER) Ithaca, *2016*

Cornell Institute for Social and Economic Research (CISER) Ithaca, *2016*

Prof. Ping Qin, Renmin University of China Beijing, *2014, 2015*

Other Experience

Data Analyst Intern, J. D. Power and Associates — Beijing, *2014*

Field Surveyor, Chinese Residential Energy Consumption Survey — Beijing & Jining, *2012-2013*

Level II Football Referee, Chinese Football Association — Beijing, *2012-2015*

HKUST MSSS Football Team, Faculty and Staff Football Team — Hong Kong, *2022* -

Software Skills

Programming: STATA, MATLAB, ArcGIS, Python, R, SAS, SPSS, ORACLE, MS SQL

Language: Mandarin Chinese (native), English (fluent), Latin (basic)

Student and RA Placements

London School of Economics and Political Science (PhD), Arizona (PhD), WashU St. Louis (PhD), HKUST Guangzhou (PhD), Georgia Tech (PhD), UC3M(PhD), UNC Chapel Hill (PhD), Tsinghua U (MFin), Imperial College (MSc) , U Chicago (MPP), HKMA, Shenzhen U (AP)